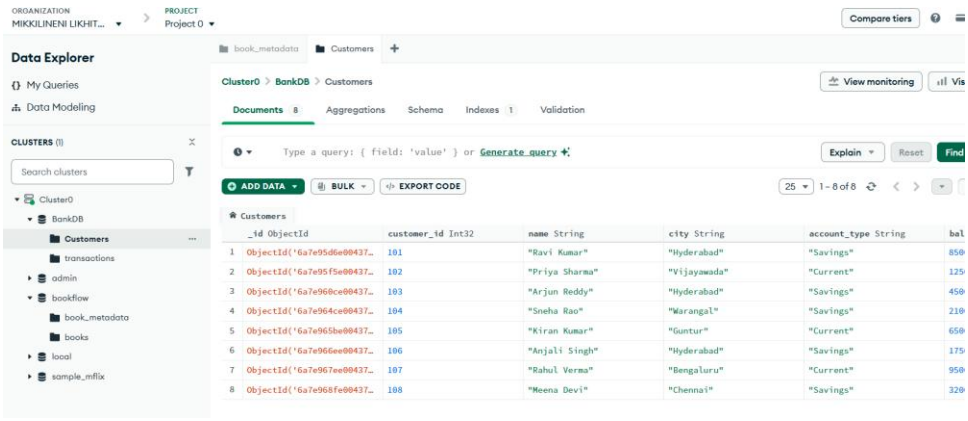
ROLL NO:2520030399

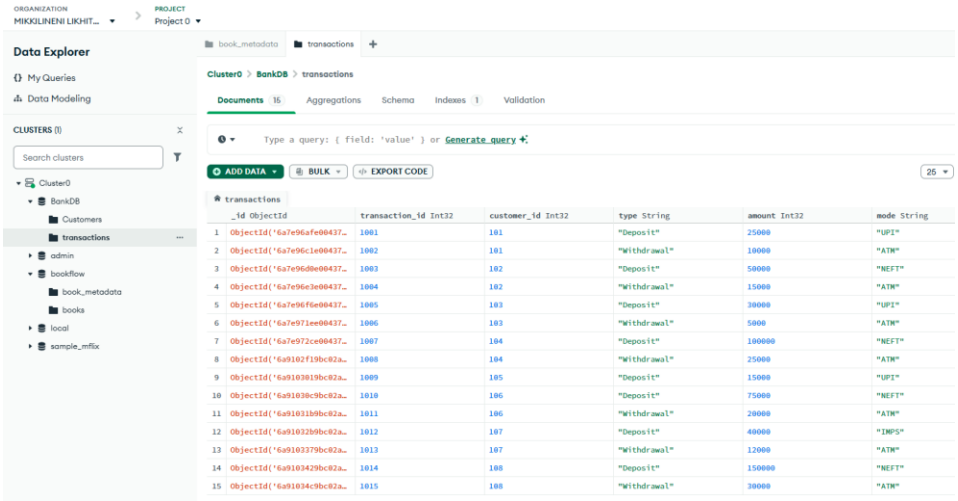
NAME: K.Harini

INSERT CUSTOMER DATA



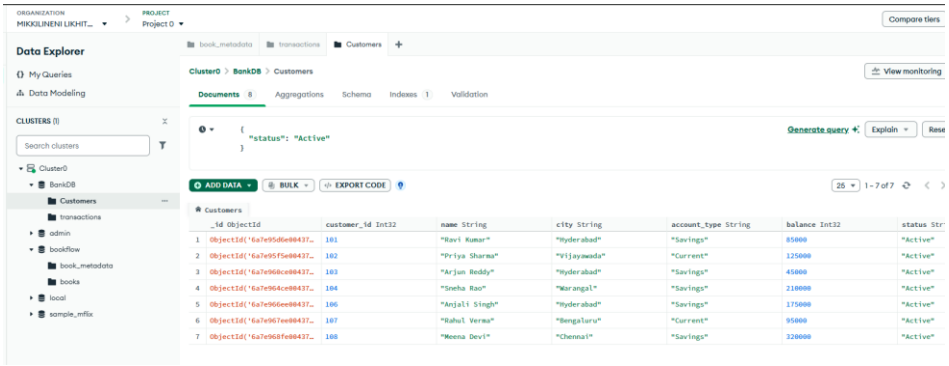
_id ObjectId	customer_id Int32	name String	city String	account_type String	bal
1 ObjectId('6a7e95d6e00437…	101	"Ravi Kumar"	"Hyderabad"	"Savings"	850
2 ObjectId('6a7e95f5e00437…	102	"Priya Sharma"	"Vijayawada"	"Current"	125
3 ObjectId('6a7e960ce00437…	103	"Arjun Reddy"	"Hyderabad"	"Savings"	450
4 ObjectId('6a7e964ce00437…	104	"Sneha Rao"	"Warangal"	"Savings"	210
5 ObjectId('6a7e965be00437…	105	"Kiran Kumar"	"Guntur"	"Current"	650
6 ObjectId('6a7e966ee00437…	106	"Anjali Singh"	"Hyderabad"	"Savings"	175
7 ObjectId('6a7e967ee00437…	107	"Rahul Verma"	"Bengaluru"	"Current"	950
8 ObjectId('6a7e968fe00437…	108	"Meena Devi"	"Chennai"	"Savings"	320

INSERT TRANSACTION DATA



_id ObjectId	transaction_id Int32	customer_id Int32	type String	amount Int32	mode String
1 ObjectId('6a7e96afe00437…	1001	101	"Deposit"	25000	"UPI"
2 ObjectId('6a7e96c1e00437…	1002	101	"Withdrawal"	10000	"ATM"
3 ObjectId('6a7e96d8e00437…	1003	102	"Deposit"	50000	"NEFT"
4 ObjectId('6a7e96e3e00437…	1004	102	"Withdrawal"	15000	"ATM"
5 ObjectId('6a7e96f6e00437…	1005	103	"Deposit"	30000	"UPI"
6 ObjectId('6a7e971ee00437…	1006	103	"Withdrawal"	5000	"ATM"
7 ObjectId('6a7e972ce00437…	1007	104	"Deposit"	100000	"NEFT"
8 ObjectId('6a9102f19bc02a…	1008	104	"Withdrawal"	25000	"ATM"
9 ObjectId('6a9103019bc02a…	1009	105	"Deposit"	15000	"UPI"
10 ObjectId('6a91030c9bc02a…	1010	106	"Deposit"	75000	"NEFT"
11 ObjectId('6a91031b9bc02a…	1011	106	"Withdrawal"	20000	"ATM"
12 ObjectId('6a91032b9bc02a…	1012	107	"Deposit"	40000	"IMPS"
13 ObjectId('6a9103379bc02a…	1013	107	"Withdrawal"	12000	"ATM"
14 ObjectId('6a9103429bc02a…	1014	108	"Deposit"	150000	"NEFT"
15 ObjectId('6a91034c9bc02a…	1015	108	"Withdrawal"	30000	"ATM"

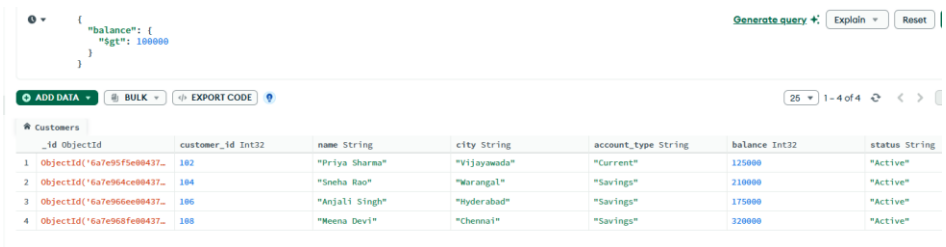
QUERY 2: FIND ACTIVE CUSTOMERS



```
{
  "status": "Active"
}
```

_id ObjectId	customer_id Int32	name String	city String	account_type String	balance Int32	status Str
1 ObjectId('6a7e95d6e00437…	101	"Ravi Kumar"	"Hyderabad"	"Savings"	85000	"Active"
2 ObjectId('6a7e95f5e00437…	102	"Priya Sharma"	"Vijayawada"	"Current"	125000	"Active"
3 ObjectId('6a7e960ce00437…	103	"Arjun Reddy"	"Hyderabad"	"Savings"	45000	"Active"
4 ObjectId('6a7e964ce00437…	104	"Sneha Rao"	"Warangal"	"Savings"	210000	"Active"
5 ObjectId('6a7e966ee00437…	106	"Anjali Singh"	"Hyderabad"	"Savings"	175000	"Active"
6 ObjectId('6a7e967ee00437…	107	"Rahul Verma"	"Bengaluru"	"Current"	95000	"Active"
7 ObjectId('6a7e968fe00437…	108	"Meena Devi"	"Chennai"	"Savings"	320000	"Active"

QUERY 3: FIND CUSTOMERS WITH BALANCE GREATER THAN 100000



```
{
  "balance": {
    "$gt": 100000
  }
}
```

_id ObjectId	customer_id Int32	name String	city String	account_type String	balance Int32	status String
1 ObjectId('6a7e95f5e00437…	102	"Priya Sharma"	"Vijayawada"	"Current"	125000	"Active"
2 ObjectId('6a7e964ce00437…	104	"Sneha Rao"	"Warangal"	"Savings"	210000	"Active"
3 ObjectId('6a7e966ee00437…	106	"Anjali Singh"	"Hyderabad"	"Savings"	175000	"Active"
4 ObjectId('6a7e968fe00437…	108	"Meena Devi"	"Chennai"	"Savings"	320000	"Active"

QUERY 4: FIND HYDERABAD CUSTOMERS:

```
{
  "city": "Hyderabad"
}
```

Generate query | Explain | Res

ADD DATA | BULK | EXPORT CODE | 25 | 1 - 3 of 3

Customers

	_id ObjectId	customer_id Int32	name String	city String	account_type String	balance Int32	status Str
1	ObjectId('6a7e95d6e00437…	101	"Ravi Kumar"	"Hyderabad"	"Savings"	85000	"Active"
2	ObjectId('6a7e960ce00437…	103	"Arjun Reddy"	"Hyderabad"	"Savings"	45000	"Active"
3	ObjectId('6a7e966ee00437…	106	"Anjali Singh"	"Hyderabad"	"Savings"	175000	"Active"

QUERY 5: FIND SAVINGS ACCOUNT CUSTOMERS:

```
{
  "account_type": "Savings"
}
```

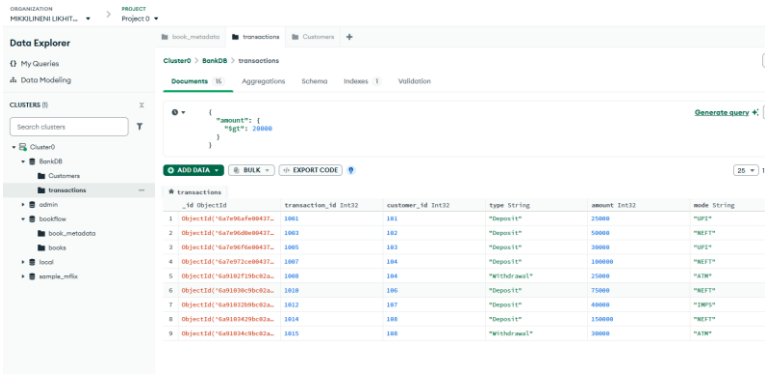
Generate query | Explain | Reset | Find

ADD DATA | BULK | EXPORT CODE | 25 | 1 - 5 of 5

Customers

	_id ObjectId	customer_id Int32	name String	city String	account_type String	balance Int32	status String
1	ObjectId('6a7e95d6e00437…	101	"Ravi Kumar"	"Hyderabad"	"Savings"	85000	"Active"
2	ObjectId('6a7e960ce00437…	103	"Arjun Reddy"	"Hyderabad"	"Savings"	45000	"Active"
3	ObjectId('6a7e964ce00437…	104	"Sneha Rao"	"Warangal"	"Savings"	210000	"Active"
4	ObjectId('6a7e966ee00437…	106	"Anjali Singh"	"Hyderabad"	"Savings"	175000	"Active"
5	ObjectId('6a7e968fe00437…	108	"Meena Devi"	"Chennai"	"Savings"	320000	"Active"

QUERY 6: FIND TRANSACTIONS GREATER THAN 20000:



QUERY 7: FIND DEPOSITS

```
{
  "type": "Deposit"
}
```

Generate query

ADD DATA | BULK | EXPORT CODE | 25

transactions

	_id ObjectId	transaction_id Int32	customer_id Int32	type String	amount Int32	mode String
1	ObjectId('6a7e96afe00437…	1001	101	"Deposit"	25000	"UPI"
2	ObjectId('6a7e96d0e00437…	1003	102	"Deposit"	50000	"NEFT"
3	ObjectId('6a7e96f6e00437…	1005	103	"Deposit"	30000	"UPI"
4	ObjectId('6a7e972ce00437…	1007	104	"Deposit"	100000	"NEFT"
5	ObjectId('6a9103019bc02a…	1009	105	"Deposit"	15000	"UPI"
6	ObjectId('6a91030c9bc02a…	1010	106	"Deposit"	75000	"NEFT"
7	ObjectId('6a91032b9bc02a…	1012	107	"Deposit"	40000	"IMPS"
8	ObjectId('6a9103429bc02a…	1014	108	"Deposit"	150000	"NEFT"

QUERY 8: FIND ATM TRANSACTIONS

```
{
  "mode": "ATM"
}
```

Generate query

ADD DATA | BULK | EXPORT CODE | 25

transactions

	_id ObjectId	transaction_id Int32	customer_id Int32	type String	amount Int32	mode String
1	ObjectId('6a7e96c1e00437…	1002	101	"Withdrawal"	10000	"ATM"
2	ObjectId('6a7e96e3e00437…	1004	102	"Withdrawal"	15000	"ATM"
3	ObjectId('6a7e971ee00437…	1006	103	"Withdrawal"	5000	"ATM"
4	ObjectId('6a9102f19bc02a…	1008	104	"Withdrawal"	25000	"ATM"
5	ObjectId('6a91031b9bc02a…	1011	106	"Withdrawal"	20000	"ATM"
6	ObjectId('6a9103379bc02a…	1013	107	"Withdrawal"	12000	"ATM"
7	ObjectId('6a91034c9bc02a…	1015	108	"Withdrawal"	30000	"ATM"

QUERY 9: FIND DEPOSITS GREATER THAN 50000

```
{
  "type": "Deposit",
  "amount": {
    "$gt": 50000
  }
}
```

Generate query | Explain | Reset

ADD DATA | BULK | EXPORT CODE

25 | 1 - 3 of 3

transactions

	_id ObjectId	transaction_id Int32	customer_id Int32	type String	amount Int32	mode String	date String
1	ObjectId('6a7e972ce00437…	1007	104	"Deposit"	100000	"NEFT"	"2026-08-04"
2	ObjectId('6a91030c9bc02a…	1010	106	"Deposit"	75000	"NEFT"	"2026-08-06"
3	ObjectId('6a9103429bc02a…	1014	108	"Deposit"	150000	"NEFT"	"2026-08-08"

QUERY 10: FIND HYDERABAD CUSTOMERS WITH HIGH BALANCE

```
{
  "city": "Hyderabad",
  "balance": {
    "$gt": 100000
  }
}
```

Generate query | Explain | Reset

ADD DATA | BULK | EXPORT CODE

25 | 1 - 1 of 1

Customers

	_id ObjectId	customer_id Int32	name String	city String	account_type String	balance Int32	status String
1	ObjectId('6a7e966ee00437…	106	"Anjali Singh"	"Hyderabad"	"Savings"	175000	"Active"

QUERY 11: FIND CUSTOMERS IN MULTIPLE CITIES

```
{
  "city": {
    "$in": [
      "Hyderabad",
      "Vijayawada",
      "Warangal"
    ]
  }
}
```

Generate query

ADD DATA | BULK | EXPORT CODE

25 | 1

Customers

	_id ObjectId	customer_id Int32	name String	city String	account_type String	balance Int32
1	ObjectId('6a7e95d6e00437…	101	"Ravi Kumar"	"Hyderabad"	"Savings"	85000
2	ObjectId('6a7e95f5e00437…	102	"Priya Sharma"	"Vijayawada"	"Current"	125000
3	ObjectId('6a7e960ce00437…	103	"Arjun Reddy"	"Hyderabad"	"Savings"	45000
4	ObjectId('6a7e964ce00437…	104	"Sneha Rao"	"Warangal"	"Savings"	210000
5	ObjectId('6a7e966ee00437…	106	"Anjali Singh"	"Hyderabad"	"Savings"	175000

QUERY 12: SORT CUSTOMERS BY BALANCE

Stage 1 | $sort

```
{
  "balance": -1
}
```

Output preview after $sort stage (Sample of 8 documents)

```
_id: ObjectId('6a7e968fe00437c4427e9e40')
customer_id : 108
name : "Meena Devi"
city : "Chennai"
account_type : "Savings"
balance : 220000
status : "Active"
```

```
_id: ObjectId('6a7e964ce00437c4427e9e38')
customer_id : 104
name : "Sneha Rao"
city : "Warangal"
account_type : "Savings"
balance : 210000
status : "Active"
```

```
_id: ObjectId('6a7e966ee00437c4427e9e3c')
customer_id : 106
name : "Anjali Singh"
city : "Hyderabad"
account_type : "Savings"
balance : 175000
status : "Active"
```

```
_id: ObjectId('6a7e95f5e00437c4427e9e34')
customer_id : 102
name : "Priya Sharma"
city : "Vijayawada"
account_type : "Current"
balance : 125000
status : "Active"
```

QUERY 13: CALCULATE TOTAL TRANSACTION AMOUNT

Stage 1 | $group

```
{
  "_id": null,
  "total_amount": {
    "$sum": "$amount"
  }
}
```

Output preview after $group stage (Sample of 1 document)

```
_id: null
total_amount : 0
```

QUERY 14: CALCULATE AVERAGE TRANSACTION AMOUNT

Stage 1 | $group

```
{
  "_id": null,
  "average_amount": {
    "$avg": "$amount"
  }
}
```

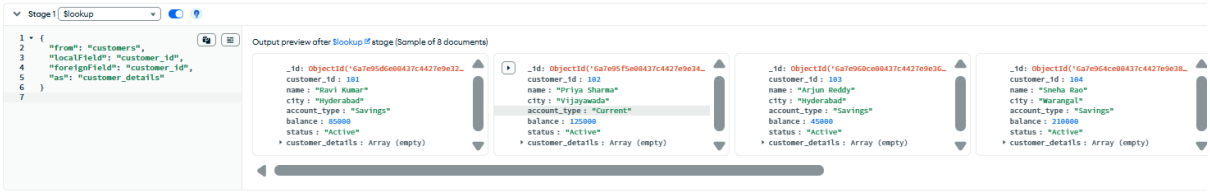
Output preview after $group stage (Sample of 1 document)

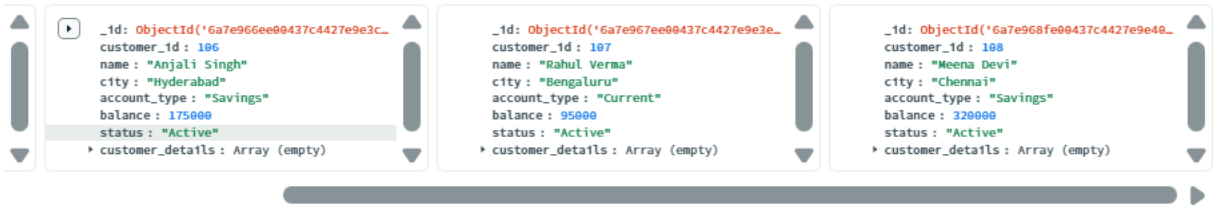
```
_id: null
average_amount : null
```

QUERY 15: TOTAL TRANSACTION AMOUNT BY TYPE



QUERY 16: CUSTOMER + TRANSACTION JOIN





QUERY 17: DISPLAY CUSTOMER NAME AND TRANSACTION AMOUNT

